YouTube Fitness Finder: API Capabilities and Build Notes

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The app

The fitness video finder is built and starts, but live search is not yet confirmed: the search call to Google failed from my sandbox with "fetch failed", most likely a network block there, so run it on your own machine to check.

- **server.js:** a dependency-free Node 18+ server that calls the YouTube API with the key from .env, so the key never reaches the browser. Results are cached for an hour to save quota.
- **public/index.html:** search box, topic chips (HIIT, yoga, kettlebell and others), sort and length filters, and a video grid with views, likes, duration and channel, plus "Load more".
- **.env:** holds YOUTUBE_API_KEY and PORT. The .gitignore keeps it out of git.

To run it, put the files in one folder with index.html inside a public subfolder, run `node server.js`, and open http://localhost:3000.

The app uses the official API, not page scraping. Scraping YouTube breaks its terms of service, and the API returns cleaner data.

Secure your key

Treat the API key as exposed, because it was pasted into a chat. In Google Cloud Console, create a new key and delete the old one.

- Restrict the new key to the YouTube Data API v3 only.
- Keep .env out of git; the supplied .gitignore already does this.
- Never put the key in browser code; the app calls the API from the server.

What the API can do

The YouTube Data API v3 reads and, with sign-in, changes YouTube content across many resource types.

- **Resources:** activities, captions, channels, comments, comment threads, playlists, playlist items, search results, subscriptions, thumbnails, videos, video categories and more. Which operations (list, insert, update, delete) work depends on the resource.
- **Search:** search.list takes a query and returns up to 50 results per page, sortable by relevance, date, view count, rating or title. It returns snippet data only, so view counts need a second videos.list call, which the app makes.
- **Public data with a key:** requests made with just an API key retrieve public data only. Private user data and write actions need OAuth 2.0.
- **Choosing fields:** the part parameter is required and picks which groups of properties come back. The fields parameter trims the response further.

Data you can get

Video data is the richest, and each part of a resource is requested separately.

Resource	Data it holds
Videos	snippet (title, description, tags, thumbnails), contentDetails (duration), statistics (views, likes, comments), status, player, topicDetails, recordingDetails
Channels	subscriber and view counts, uploads playlist, featured sections
Comments and comment threads	comment text, likes, replies
Playlists and playlist items	every video in a playlist
Video categories and regions	category lists and region-specific filtering

Quota limits

The default quota is 10,000 units per day, which allows about 99 searches a day at 101 units each (search plus the stats lookup).

Operation	Cost (units)
List read (videos, channels, playlists)	1

Operation	Cost (units)
Search request	100
Write (create, update, delete)	50

At the limit the API returns a 403 quotaExceeded error until the quota resets. Extra quota can be requested from Google. The pages I found disagree on search cost: one Google overview says 100 units and another Google overview variant says 1. The app assumes 100, the safer figure; check the Quotas page in your Google Cloud Console for the real number.

Apps you could build

- **Fitness trend tracker:** save daily snapshots of view counts per topic to see which workout types are growing.
- **Creator analytics:** compare fitness channels by upload frequency, views per video and engagement rate (likes plus comments divided by views).
- **Curated workout library:** filter by duration, level and equipment, with saved playlists.
- **Content-gap finder:** find topics with high demand (many views) but few quality videos.
- **Comment insights:** summarize what viewers ask for under popular videos.
- **Sponsorship scouting:** rank smaller creators by engagement.
- **Dashboards:** send the data to Supabase or a spreadsheet for charts.

Documentation and sources

The main reference is the YouTube Data API overview, which covers setup, resources, part and fields, and quota, and links to a reference page for each resource and method. The facts above come from web search results, not full page reads.

- YouTube Data API Overview
- Understanding the YouTube Data API v3 Quota System
- YouTube Search API guide (OutlierKit)